

Question	Answer	Mark	Mark scheme	Additional guidance
23	Proof (supported)	M1	for using the sine rule on triangle ABD or on triangle ADC, to involve sides AB, BD, AC, or DC eg $\frac{AB}{\sin ADB} = \frac{BD}{\sin x}$ oe or $\frac{AC}{\sin ADC} = \frac{DC}{\sin x}$ oe OR for an expression for the area of triangle ABD or for the area of triangle ADC eg $\frac{1}{2} AB AD \sin x$ or $\frac{1}{2} AD AC \sin x$ or $\frac{1}{2} h BD$ or $\frac{1}{2} h DC$	Accept extra letters eg y shown on diagram for any angle used
		M1	for using the sine rule on both triangle ABD and on triangle ADC, to involve sides AB, BD, AC, or DC eg $\frac{AB}{\sin ADB} = \frac{BD}{\sin x}$ oe and $\frac{AC}{\sin ADC} = \frac{DC}{\sin x}$ oe OR for two expressions for the area of either triangle ABD or for triangle ADC eg $\frac{1}{2} AB AD \sin x$ and $\frac{1}{2} h BD$ or $\frac{1}{2} AD AC \sin x$ and $\frac{1}{2} h DC$	
		M1	for stating or showing $\sin ADB = \sin ADC$, eg $\sin y = \sin (180 - y)$ OR for using two expressions to form an equation eg $\frac{\frac{1}{2} AB AD \sin x}{\frac{1}{2} AD AC \sin x} = \frac{\frac{1}{2} h BD}{\frac{1}{2} h DC}$ oe	
		C1	for a full method to arrive at the given answer	